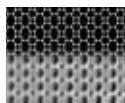




Improved solar cell efficiency

New ultrathin metal electrodes used to create semitransparent perovskite solar cells that are highly efficient. <http://digit.in/jun21-63>



New carbon material discovered

A new carbon network, like graphene, but made up of squares, hexagons, and octagons forming an ordered lattice. <http://digit.in/jun21-64>

the universe, they should be bursting into photons in the x-ray band, that should be picked up by astronomical instruments on Earth. To test the theory MIT researchers studied the star Betelgeuse, which is one of those stars that may burst into a supernova at any time. The researchers did not find the axions that they were looking for, but that is actually very good news. This is because by not finding the axions, the physicists were able to put better constraints on the characteristics of the particle, allowing them to fine tune their search. The conversion of axions to photons depends on something known as the Primakoff effect, a theoretical coupling between axions and photons, which predicts that when photons are subjected to intense magnetic fields, they are squeezed into becoming axions. These intense magnetic fields exist at the hearts of the stars about to explode. In the light coming from a young energetic star such as the Sun, the axions could well hide in the x-rays, but an old dying star such as Betelgeuse don't stream out x-rays according to current astrophysical models, but should be veritable factories of axions. The team's findings was that they would have to look for the axions in the bands where they would be more difficult to spot. Physicists are waiting for a star such as Betelgeuse to actually explode, and if they do not see the signature of the axions in the super-

nova also, then that would put further constraints on the coupling between axions and photons.

A team from the University of Minnesota has actually found signals that look like they could be axions streaming out of a neutron star. Raymond Co, an author on the paper and postdoctoral researcher in the University of Minnesota's School of Physics and Astronomy says, "Finding axions has been one of the major efforts in high-energy particle physics, both in theory and in experiments. We think axions could exist, but we haven't discovered them yet. You can think of axions as ghost particles. They can be anywhere in the universe, but they don't interact strongly with us so we don't have any observations of them yet." The team noticed that some nearby neutron stars were emitting more x-rays than they should, and that the difference in the expected observations could be explained by axions. The neutron star itself provides the magnetic field necessary to convert the axions into photons. Co says, "We're not claiming that we've made the discovery of the axion yet, but we're saying that the extra x-ray photons can be explained by axions. It is an exciting discovery of the excess in the x-ray photons, and it's an exciting possibility that's already consistent with our interpretation of axions."

The suspected Primakoff effect is used in Earth based detectors to hunt for the elusive axions as well. In May, the The Relic Axion Detector Experimental Setup (RADES) went live at CERN, in an attempt at detecting axions from the halo of the Milky Way. The detectors use what are known as resonating cavities, where axions can be trapped and intense magnetic fields applied to them, to see if they morph into photons. So far there was a limit to the detectors, as detecting larger mass axions required cavities

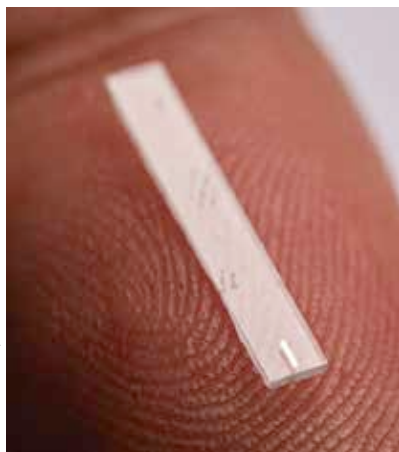
of smaller volumes resonating at higher frequencies, but the size of the cavities greatly reduced the chances of interacting with an axion. The physicists got around the problem by dividing the cavities into smaller subcavities, that all resonate at higher frequencies without reducing the volume for detection, providing the RADES instrument. The RADES instrument is an upgrade to the CERN Axion Solar Telescope (CAST) experiment, which previously searched for axions from the Sun.



The XMM-Newton telescope detected the anomalous amount of x-rays from nearby neutron stars

RADES scientist Sergio Arguedas Cuendis says, "This result is a significant first step in the direct search for axions using dipole magnets. And as far as axion searches go, it's one of the most stringent limits ever set for axions with masses above 25 μeV ."

Researchers from Fermilab have used another innovative method of getting around the volume of detection problem, by using superconducting microwave cavities to accumulate the dark matter signal. The best thing about this new qubit based device is that the photon remains trapped in the cavity holds the photon and provides continuous data over 500 microseconds of its existence. The approach has the advantage of eliminating false positive detections, reduce the noise, and increase the sensitivity of the detectors. Despite all the efforts by the best detectors in the world, axions remain at this point, elusive. However, a global hunt is on, and there remains very few places for this particle to hide. 



A cubit similar to this one was modified into a dark matter detector